Tourism Data Analysis & Travel Trend Visualization

An Empirical Study on Post-Pandemic Visitor Patterns & Market Insights

Prepared for Academic Assessment | Department of Computer Science & Engineering

TOURISM DATA ANALYSIS

Project Overview & Objectives

Core Objectives

- **Shift Identification:** Analyze post-pandemic mobility data to identify structural shifts in travel behavior.
- **Demographic Segmentation:** Evaluate changing preferences across age groups, travel durations, and budget tiers.
- **Seasonal Footfall Modeling:** Examine occupancy fluctuations across peak, shoulder, and off-peak months.
- **Decision Support:** Deliver data-driven recommendations for destination management and dynamic pricing.

Scope & Target Metrics

- **Primary Data Sources:** Aviation mobility records, regional hotel occupancy datasets, and booking portal metrics.
- **Analytical Methods:** Exploratory Data Analysis (EDA), seasonal time-series decomposition, and footfall heatmapping.
- **Primary Output:** Visual dashboards and statistical summaries for regional tourism planning.

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Dataset & Data Pipeline Methodology

01. Data Ingestion

Gathering raw mobility records, flight booking feeds, and hotel occupancy statistics from regional archives.

02. Preprocessing & Cleaning

Handling null records, removing duplicate entries, and normalizing seasonal variables across timeline intervals.

03. Exploratory Analysis

Statistical aggregation, correlation analysis between pricing and footfall, and sentiment clustering.

04. Visualization & Reporting

Generating metric dashboards, occupancy trend charts, and strategic policy recommendations.

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Trend 1: Rise of Experiential & Micro-Tourism

38% Increase

Short Getaways vs. Long Stays

Travelers increasingly prefer frequent 2-3 day weekend trips over traditional extended two-week vacations.

2.4x Growth

Heritage & Eco-Tourism

High growth in demand for authentic cultural immersion, local food trails, and eco-certified stays.

+42% Footfall

Secondary Destinations

Offbeat and non-metropolitan destinations are recording higher year-over-year percentage growth than saturated hubs.

TOURISM DATA ANALYSIS

Trend 2: Seasonal Footfall & Occupancy Analysis

Key Seasonal Observations

- **Shoulder Season Expansion:** Monsoon and off-peak quarters saw a 18% surge in mid-budget bookings.
- **Peak Smoothing:** Extreme summer weather is spreading traditional summer demand into early spring and autumn months.
- **Revenue Opportunity:** Dynamic off-peak packaging can optimize year-round facility utilization for hotel operators.

Quarter / Period	Avg. Occupancy Rate	YoY Growth
Q1 (Jan - Mar)	68%	+12.4%
Q2 (Apr - Jun)	84%	+5.1%
Q3 (Jul - Sep)	59%	+18.2%
Q4 (Oct - Dec)	91%	+8.7%

Strategic Recommendations & Summary

1. Dynamic Pricing Architecture

Implement algorithmic price elasticity models for municipal transit, cultural entry passes, and regional accommodation during shoulder seasons.

2. Heritage Corridor Infrastructure

Invest in last-mile transport infrastructure connecting major transit hubs to emerging secondary heritage and eco-tourism sites.

3. Off-Peak Target Marketing

Focus digital marketing campaigns around weekend micro-vacations and cultural experiences during traditionally low-occupancy months.